

more **for** loops.

Week **6** | Lecture **2** (6.2)

Upcoming

- Term Test 1, Thursday at 7:00 pm.
- Online Term Test Review, Recording Posted.
- Reflection 6 Released Friday, 6:00 pm.
- Tutorial (Online), Office Hour sessions running all week.

if nothing else, write **#cleancode**.

This Week's Content

- Lecture 6.1
 - for loops
- Lecture 6.2
 - for loops on indices, nested loops
- Lecture 6.3
 - Design Problem: Cryptography

for loops

- A **for** loop starts with the keyword **for**.

```
name = 'Sebastian'
```

```
for character in name:  
    print(character)
```

for loops

- Next, we provide the name of one of more variables.
- We have called the variable `character`, but you can call it whatever you like as long as it follows rules for naming a variable.

```
name = 'Sebastian'
```

```
for character in name:  
    print(character)
```

```
for item1, item2 in iterable:  
    do something.
```

for loops

- Our variable `character` will be bound to each of the items in the sequence in turn.

```
name = 'Sebastian'
```

```
for character in name:  
    print(character)
```

for loops

- Specify what the values are in.
- What is the iterable?
- An iterable is an object that can be iterated over.
- Strings are iterable (we know these from last week).
- Lists (**next week**) are iterable.

```
name = 'Sebastian'
```

```
for character in name:  
    print(character)
```

for loops

- As with the `while` loop, the `for` loop statement ends with a colon.
- This is how Python knows you are going to create a new block of code.

```
name = 'Sebastian'
```

```
for character in name:  
    print(character)
```

for loops

- Indenting four spaces tells Python what lines of code are in that block you want to repeated.

```
name = 'Sebastian'
```

```
for character in name:  
    print(character)
```



Indent

Breakout Session 1

- We want to do some analysis of Dean Yip's Tweets.
- Before we can do this, we'll need to make the tweet all lower case and replace all the punctuations with white space.
- 'impact... Exciting' → 'impact exciting'



Chris Yip @UofTEngDean · Oct 23

Replying to @UofTEngDean

great to hear from Prof. Bussman, Chair of @uoftmie about all the stuff that MechE do - amazing breadth of impact... exciting stuff across so many domains

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1. Breakout Session 1

Today's Content

- Looping through indices with a **for** loop.
- Nested **for** loops.

Looping Through Indices

- Last lecture we saw that we can use while loops to loop over the indices of a string.
- Then we saw that a for-loop requires less code but it iterates over the values, not the indices.

while

```
i = 0
```

```
while i < len(chrome_4):  
    print(i, chrome_4[i])  
    i += 1
```

for

```
for character in chrome_4:  
    print(character)
```

Looping Through Indices

- Can we use a for loop to loop over indices?

```
while
i = 0
while i < len(chrome_4):
    print(i, chrome_4[i])
    i += 1
```

```
for
for character in chrome_4:
    print(character)
```

Looping on a `range()`

- Python has a built-in function called `range()` that can be used to generate a sequence of numbers. The general syntax of range is as follows:

`range(start, stop, step)`

- Similar to the string slicing syntax:
 - The stop value is not included in the sequence of numbers generated.
 - Can omit start and step which will result in default values being used. `range(n) → range(0, n, 1)`

Looping on a `range()`

- `range()` is typically used in a for loop to iterate over a sequence of numbers.
- `range()` is an iterable.

This thing has to be an iterable.



```
for i in range(5):  
    print(i)
```

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2. Using range()

Example 1

- Add up all the even numbers between 1 and 100 using a `for` loop and `range()`.
- $2 + 4 + \dots + 96 + 98 + 100$

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3. Example 1

Breakout Session 2

- Write a function that returns the number of times that a character and the next character are the same.
- If you have a bug in a loop, with probability ~ 1 its an off-by-one index error.

```
count_adjacent_repeats('abccdeffggh')
```

```
>>> 3
```

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4. Breakout Session 2

Nested **for** Loops

- The bodies of loops can contain any statement, including other loops!
- When this occurs, it is known as a nested loop.

```
for item in iterable:  
    do something.
```

```
for i in range(10, 13):  
    for j in range(1, 5):  
        print(i, j)
```

Output

```
10, 1  
10, 2  
10, 3
```

...

Nested **for** Loops

- The bodies of loops can contain any statement, including other loops!
- When this occurs, it is known as a nested loop.

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5. Nested for Loops

Turtles

- **Turtle** is a pre-installed Python library that enables users to create pictures and shapes by providing them with a virtual canvas.

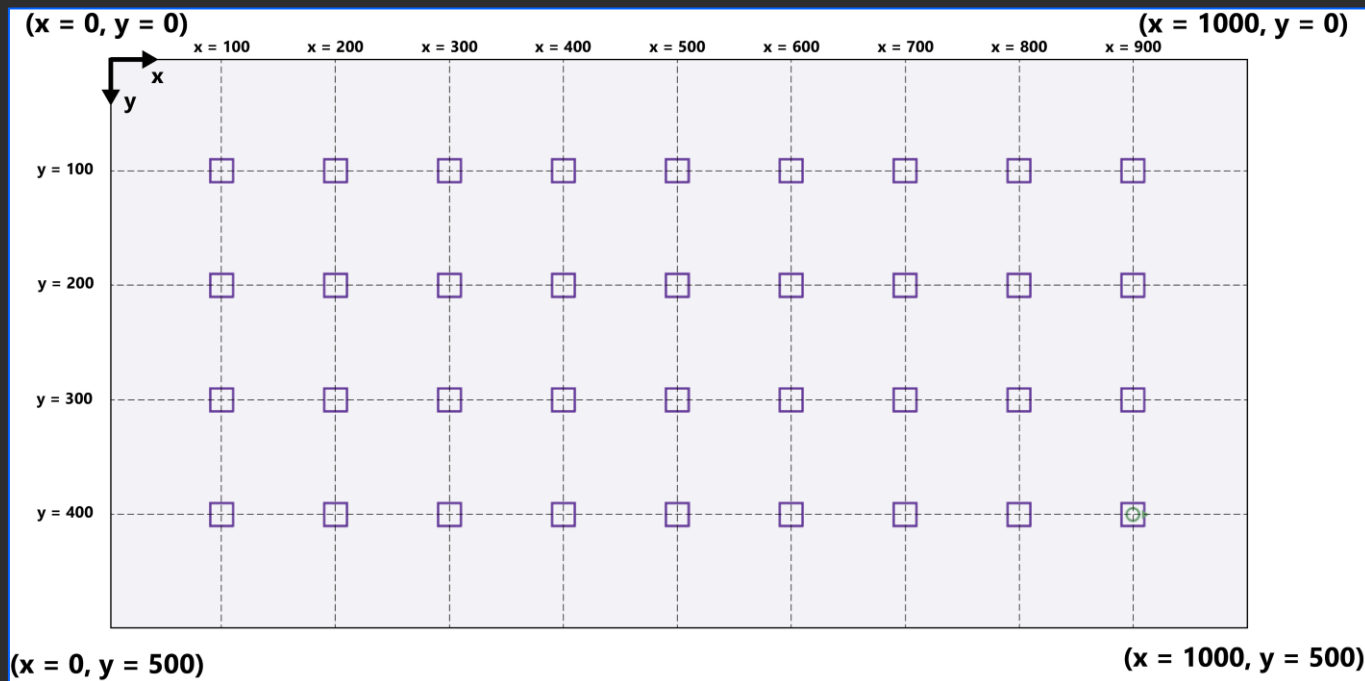


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6. Turtles

Draw A Grid

- Use Turtle and nested for loops to draw a grid.

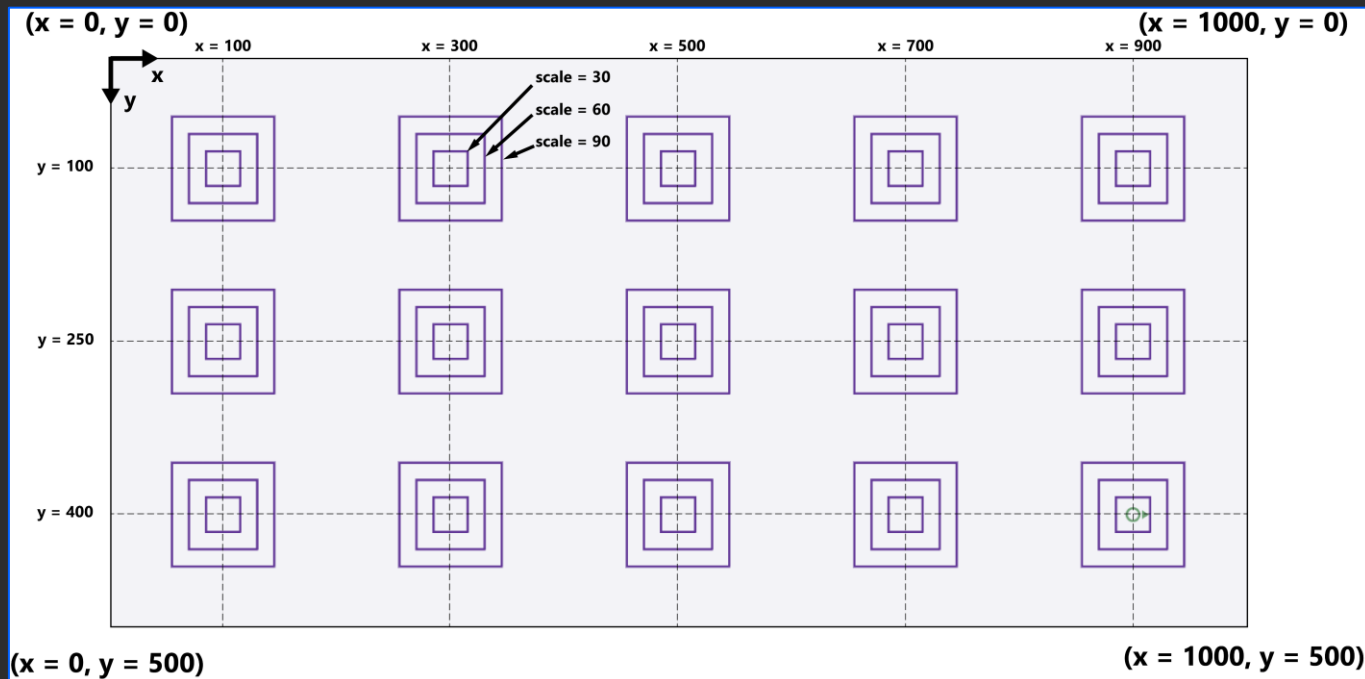


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7. Grid

Breakout Session 3

- Use Turtle and nested for loops to draw the pattern below.

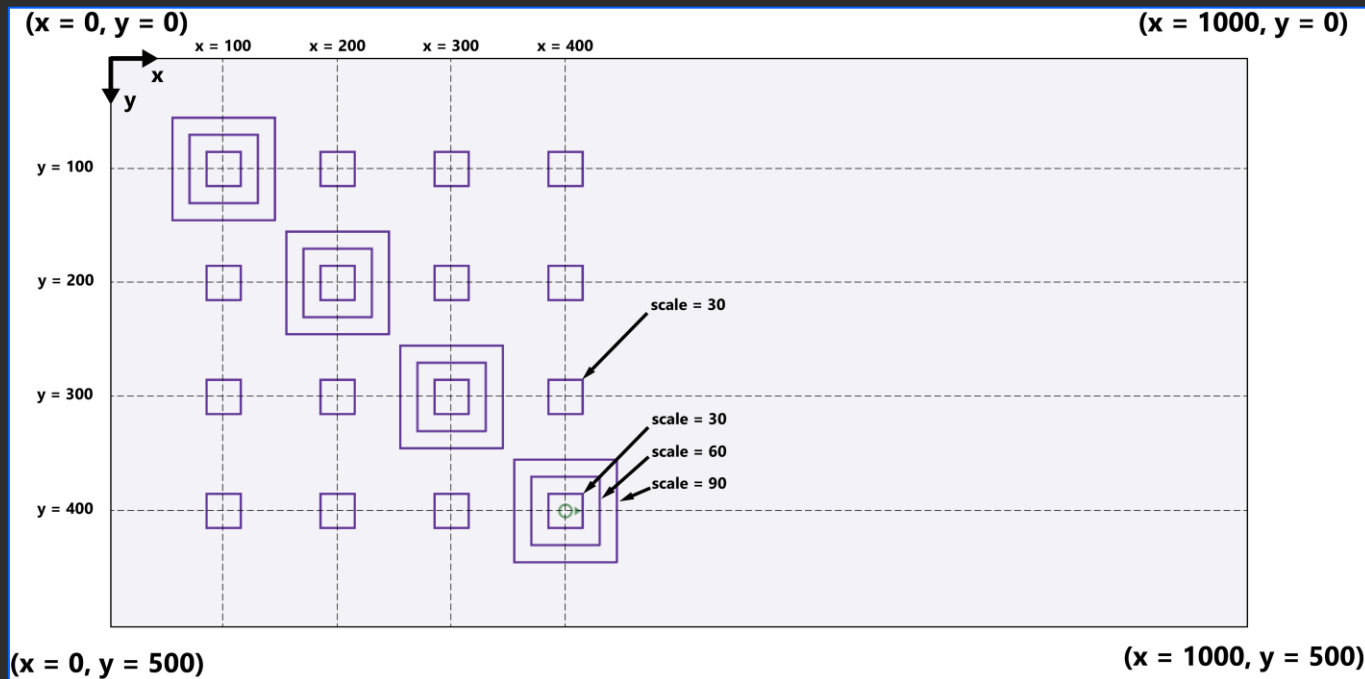


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8. Breakout Session 3

Breakout Session 4

- Use Turtle and nested for loops to draw the pattern below.



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9. Breakout Session 4

PRACTICE!

more **for** loops.

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